

Matrices, Determinants and Systems of linear equations

Operations of determinants:

- 1) $L_i \leftrightarrow L_j$ (swap)
- 2) $L_i \rightarrow \alpha L_i$ (multiplying by a scalar)
- 3) $L_i \rightarrow \alpha L_j + L_i$ (mult. a diff. line then add)

$$\text{ex. 1) } D = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 1 \\ -1 & 4 & -1 & 0 \\ 0 & 2 & 1 & 1 \end{vmatrix} \xrightarrow{\substack{L_2 - 2L_1 \\ L_3 + L_1}} \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -5 \\ 0 & 6 & -2 & 3 \\ 0 & 2 & 1 & 1 \end{vmatrix}$$

$$= 1 \cdot \begin{vmatrix} -5 & 5 & -5 \\ 6 & -2 & 3 \\ 2 & 1 & 1 \end{vmatrix} \cdot (-1)^{1+1} = \begin{vmatrix} -5 & 5 & -5 \\ 6 & -2 & 3 \\ 2 & 1 & 1 \end{vmatrix} = 5 \cdot \begin{vmatrix} -1 & 1 & -1 \\ 6 & -2 & 3 \\ 2 & 1 & 1 \end{vmatrix}$$

$$\xrightarrow{\substack{C_1 + C_2 \\ C_1 + C_3}} 5 \cdot \begin{vmatrix} 0 & 1 & 0 \\ 4 & -2 & 1 \\ 3 & 3 & 2 \end{vmatrix} = 1 \cdot (-1)^{1+2} \cdot \begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix} \cdot 5 = -5 \cdot 5 = -25$$

$$2) D = \begin{vmatrix} -2-a & -1 & 1 \\ 5 & -1-a & 4 \\ 5 & 1 & 2-a \end{vmatrix} \xrightarrow{C_3 + C_1} \begin{vmatrix} -2-a & -1 & 0 \\ 5 & -1-a & 3-a \\ 5 & 1 & 3-a \end{vmatrix}$$

$$= (3-a) \begin{vmatrix} -2-a & -1 & 0 \\ 5 & -1-a & 1 \\ 5 & 1 & 1 \end{vmatrix} \xrightarrow{(3-a) \cdot L_2 - L_3} \begin{vmatrix} -2-a & -1 & 0 \\ 0 & -2-a & 0 \\ 5 & 1 & 1 \end{vmatrix}$$

$$= (3-a) \cdot 1 \cdot (-1)^{3+3} \cdot \begin{vmatrix} -2-a & 1 \\ 0 & -2-a \end{vmatrix} = (a-3)(a+2) \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}$$

$$= (a-3)(a+2)$$

$$D(a) = 0 \Leftrightarrow (a-3)(a+2) = 0 \Rightarrow a_1 = 3$$

$$a_2 = -2$$

$$3) A = \begin{pmatrix} 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{L_3 - L_1} \begin{pmatrix} 2 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 2 & -1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

$$\underbrace{L_3 - L_2}$$

$$\underbrace{L_4 - L_2}$$

$$\Rightarrow \text{rank}(A) = 3$$

$$4) B = \begin{pmatrix} 2 & 1 & 3 & -1 \\ 3 & -1 & 2 & 0 \\ 1 & 3 & 4 & -2 \\ 4 & -3 & 1 & 1 \end{pmatrix} \xrightarrow{L_3 \leftrightarrow L_4} \begin{pmatrix} 1 & 3 & 4 & -2 \\ 3 & -1 & 2 & 0 \\ 4 & -3 & 1 & 1 \\ 2 & 1 & 3 & -1 \end{pmatrix}$$

$$\underbrace{L_2 - 3L_1}$$

$$L_3 - 2L_1$$

$$L_4 - 4L_1$$

$$\begin{pmatrix} 1 & 3 & 4 & -2 \\ 0 & -10 & -10 & 6 \\ 0 & -5 & -5 & 3 \\ 0 & -15 & -15 & 9 \end{pmatrix}$$

$$\underbrace{L_2 \leftrightarrow L_3}$$

$$\begin{pmatrix} 1 & 3 & 4 & -2 \\ 0 & -5 & -5 & 3 \\ 0 & -10 & -10 & 6 \\ 0 & -15 & -15 & 9 \end{pmatrix}$$

$$\underbrace{L_3 - 2L_2}$$

$$\underbrace{L_4 - 3L_2}$$

$$\begin{pmatrix} 1 & 3 & 4 & -2 \\ 0 & -5 & -5 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \text{rank}(B) = 2$$

$$[A|I_4] = \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 2 & -1 & -1 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} L_3 + L_1 \\ L_4 - 2L_1 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -3 & -2 & 0 & 0 & 1 \end{array} \right] \quad L_4 + L_2 \quad \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -4 & -2 & 1 & 1 & 1 \end{array} \right]$$

$$L_4 \leftrightarrow L_3 \quad \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & -4 & -2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array} \right] \quad \begin{array}{l} L_3 + 4L_4 \\ L_2 + L_3 \\ L_1 - 2L_2 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & -1 & 3 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array} \right] \quad \text{Factor 2} \quad \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & -1 & 3 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array} \right]$$

$$L_2 - 3L_3 \quad \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & -1 & 0 & 0 & -2 & -\frac{1}{2} & -5 & -\frac{3}{2} \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array} \right] \quad \text{Factor } (-1)$$

$$L_1 + L_2 \quad \left[\begin{array}{cccc|cccc} 1 & -1 & 0 & 0 & -1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 0 & 2 & \frac{1}{2} & 5 & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array} \right] \quad \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & \frac{1}{2} & 3 & \frac{3}{2} \\ 0 & 1 & 0 & 0 & 2 & \frac{1}{2} & 5 & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 1 & \frac{1}{2} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array} \right]$$

$\underbrace{\quad}_{I_4} \quad \underbrace{\quad}_{I^{-1}}$